CHANNEL ALLOCATION & MULTIPLE ACCESS PROTOCOLS

1. Channel Allocation

Channel allocation is a process in which a single channel is divided and allotted to multiple users in order to carry out user-specific tasks.

2. Static Channel Allocation

With a static channel allocation approach, the channel's capacity is essentially divided into fixed portions. Each user is then allocated a portion.

For N competing users, the bandwidth is divided into N channels using frequency division multiplexing (FDM) and time division multiplexing (TDM). Either a fixed frequency or fixed time slot is allocated to each user.

3. Dynamic Channel Allocation

With a dynamic channel allocation approach, the allocation of the channel changes based on the traffic generated by the users.

If the traffic increases, more channels are allocated to the users; otherwise fewer channels are allocated to the users.

A dynamic channel allocation tries to get better utilization and lower delay on a channel when the traffic is unpredictable.

4. Possible Assumptions of Dynamic Channel Allocation in LANs and MANs

1. Station model: N stations independently produce frames.

2. Single-channel assumption: All stations are equivalent and can send and receive on that channel.

3. Collision assumption: If two frames overlap in time, then collision occurs. Any collision is an error, and both frames must be retransmitted.

4. Continuous Time: A frame transmission can begin at any instant.

Slotted Time: Time is slotted or divided into discrete intervals (slots). A slot may contain 0, 1, or more frames: 0 = idle slot; 1 = successful transmission; more than 1 = collision.

5. Carrier Sense: (a) A station can sense if a channel is busy; no station will attempt to use it until it goes idle. (b) No carrier sense: A station can't sense the channel before trying to use it. They just go ahead and transmit.

5. Multiple Access Protocols

Multiple access protocols are needed when there is no dedicated link and multiple stations can access the channel simultaneously.

Hence, multiple access protocols are designed to decrease collision and avoid crosstalk.

6. Types of Multiple Access Protocols

Multiple access protocols are classified into: Random access protocols, Controlled access protocols, and Channelization protocols.

7. Random Access Protocols

In this, all stations have same superiority; that is, no station has more priority than another station. Any station can send data depending on the medium's state (idle or busy).

In a random access method, each station has the right to the medium without being controlled by any other station.

If more than one station tries to send, there is an access conflict and the frames will be either destroyed or modified.

8. Random Access Procedure

To avoid access conflict, each station follows a procedure:

• When can the station access the medium?

• What can the station do if the medium is busy?

• How can the station determine the success or failure of the transmission?

• What can the station do if there is an access conflict?

9. Controlled Access Protocols

In controlled access, the stations consult one another to find which station has the right to send.

A station cannot send unless it has been authorized by other stations.

Types: Reservation, Polling, Token Passing.

10. Channelization Protocols

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code between different stations.

Types: FDMA, TDMA, CDMA.